

Formulation of hybrid nodal solver based on directional effect of wave propagation in a cell-centered Lagrangian scheme

Chunyuan Xu^{*}, Zhijun Shen, Qinghong Zeng

Institute of Applied Physics and Computational Mathematics, Beijing 100091, China

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ABSTRACT

In this work, a hybrid nodal solver is constructed to incorporate the directional effect of wave propagation in a cell-centered Lagrangian scheme. First, the direction of wave is determined via an assumption based on the Rankine–Hugoniot condition. Next, two different approximation methods are used to calculate the velocity jump and numerical fluxes. Correspondingly, a hybridization strategy is proposed to formulate a hybrid nodal solver. It is shown that the developed nodal solver can replicate two well-known ones, and its effectiveness is shown in numerical experiments. Finally, an adaptive hybridization method based on vorticity is proposed. The accuracy and robustness of the adaptive method is assessed in various tests.

1. Introduction

Lagrangian methods are frequently used for the numerical simulation of high-speed compressible fluid flows and the propagation of shock waves. These methods offer the advantage of automatically and promptly tracking multi-material interfaces by simultaneously updating the moving mesh and the local fluid. Typically, the corresponding mathematical model is described by the Euler equations, which can be solved using staggered grid schemes. In this paper, we consider instead another type of cell-centered grid scheme that solves the Euler equations in a conservative form. Such schemes commonly employ a Riemann solver to assess the numerical flux at cell interfaces.

In the Lagrangian frame, early work on Riemann solvers did not consider mesh motion properly. For instance, the well-known CAVEAT code [1] developed in the 1980s solves an interface Riemann problem across each edge to obtain the numerical flux and edge velocity. The node velocity is then computed from the edge velocity using a least-squares procedure. However, this method has the disadvantage of not satisfying the geometrical conservation law (GCL). To address this issue, Després and Mazeran [2] developed the GLACE scheme, which incorporates a nodal solver. It has been shown that the nodal solver is consistent with the mesh motion as it computes the node velocity and numerical flux simultaneously. However, the GLACE scheme is found to be sensitive to large grid aspect ratios. In an effort to resolve this aspect-ratio problem, Maire et al. [3] proposed a new scheme called EUCCLHYD, which introduces four pressures on each edge. Through theoretical proof of the 1D problem, the EUCCLHYD scheme was shown to be affected less by the grid aspect ratio than is the GLACE scheme.

Since the pioneering work of the GLACE and EUCCLHYD schemes, the method of nodal solver has gained significant attention [4–13]. Further developments of nodal solvers have been reported in various areas, such as 3D generalization [14], ALE extension [15], genuinely multidimensional extension [16], weak entropy condition with low dissipation [17], positivity-preserving and entropy stable [18], reduction of dependence on the edge length [19], and so on.

Despite the numerous studies conducted on nodal solvers, their directional impact on wave propagation has received limited investigation. Most existing nodal solvers assume that the wave spreads along the normal direction of the interface, but in reality there is a high likelihood that it can spread in any arbitrary orientation. However, there is an exception to this trend in the nodal solver developed by Burton et al. [20], who consider the presence of oblique wave patterns by introducing a tensor dissipation model. Subsequently, the numerical flux resulting from this condition is evaluated along the direction of the velocity difference, leading to improvements in stability and mesh robustness. Their findings demonstrate the advantages of incorporating a more physically relevant direction for dissipation. However, note that as constructed, the tensor dissipation model in their study involves a generic direction, while its implementation only considers the direction of the velocity difference.

In the present work, the wave propagation direction is identified via a basic assumption based on the Rankine–Hugoniot (RH) condition, then a proper approximation of the velocity field is used to formulate a hybrid nodal solver. Derived in a general way, this hybrid nodal solver is shown to replicate two typical ones [4,20]. A flexible hybridization

^{*} Corresponding author.

E-mail address: xcyuan@mail.ustc.edu.cn (C. Xu).

strategy is proposed in order to obtain high accuracy and robustness, and finally the performance of the flexible hybridization strategy is shown in numerical experiments.

The rest of this paper is organized as follows. We summarize the node-based discretization method in Section 2, describe the construction of the hybrid nodal solver in Section 3, and introduce the implementation of the Lagrangian scheme in Section 4. In Section 5, we report numerical experiments that show the applicability of the hybrid nodal solver, then we propose an adaptive hybridization strategy in Section 6 and verify its performance in Section 7. Finally, we conclude in Section 8.

2. Finite-volume discretization

2.1. Governing equations

In the Lagrangian framework, the integral form of the Euler equations in 2D pseudo-Cartesian geometry is as follows:

$$\frac{d}{dt} \int_{V(t)} \rho dV = 0, \quad (1a)$$

$$\frac{d}{dt} \int_{V(t)} dV - \int_{V(t)} \nabla \cdot \mathbf{u} dV = 0, \quad (1b)$$

$$\frac{d}{dt} \int_{V(t)} \rho \mathbf{u} dV + \int_{V(t)} \nabla P dV = 0, \quad (1c)$$

$$\frac{d}{dt} \int_{V(t)} \rho E dV - \int_{V(t)} \nabla \cdot (P\mathbf{u}) dV = 0, \quad (1d)$$

where d/dt denotes the material time derivative, ρ is the density, $\mathbf{u}$ is the velocity, P is the pressure, and E is the specific total energy. These governing equations are completed by adding an equation of state. For simplicity, the equation of state of an ideal gas is considered:

$$P = (\gamma - 1)\rho\varepsilon, \quad (2)$$

where $\varepsilon = E - |\mathbf{u}|^2/2$ is the specific internal energy. The trajectory equation is written as:

$$\frac{d\mathbf{X}}{dt} = \mathbf{u}(\mathbf{X}(t), t), \quad \mathbf{X}(0) = \mathbf{X}_0. \quad (3)$$

2.2. Node-based discretization

Many Lagrangian schemes adopt a node-based discretization of numerical fluxes. It has been demonstrated in [4] that the node-based discretization has the advantage of satisfying the GCL, which means that the fluid volume computed from Eq. (1b) is equal to the mesh volume computed from the trajectory equation Eq. (3). In this paper, we will mostly follow the notations in Ref. [4]. As shown in Fig. 1, a polygonal cell is assigned a unique index c and is denoted Ω_c . Each vertex of the mesh is assigned a unique index p . The set of cells surrounding a particular vertex p is denoted $\mathcal{C}(p)$, and the set of vertices of cell c is denoted $\mathcal{P}(c)$. The counterclockwise-ordered vertices p^- , p , p^+ are marked. The quantities p_1 and p_2 are defined as the midpoints of pp^+ and pp^- . $l_{pp_1}^c$ and $l_{pp_2}^c$ denote the lengths of the half edges pp_1 and pp_2 , and $\mathbf{n}_{pp_1}^c$ and $\mathbf{n}_{pp_2}^c$ denote the unit outward normals of the half edges pp_1 and pp_2 . To represent the subscripts in a simple manner, the following notations are introduced:

$$l_{\bar{p}}^c = l_{pp_1}, \quad l_{\underline{p}}^c = l_{pp_2}, \quad \mathbf{n}_{\bar{p}}^c = \mathbf{n}_{pp_1}, \quad \mathbf{n}_{\underline{p}}^c = \mathbf{n}_{pp_2}, \quad (4)$$

here the subscripts $\bar{p}$ and $\underline{p}$ are used to represent the elements related to half edges pp_1 and pp_2 . Two nodal pressures $\pi_{\bar{p}}^c$ and $\pi_{\underline{p}}^c$ are introduced with respect to the half edges pp_1 and pp_2 . With the above nodal notations, the semi-discrete finite volume scheme of the governing equations writes [4]:

$$m_c \frac{d}{dt} \left(\frac{1}{\rho_c} \right) - \sum_{p \in \mathcal{P}(c)} \left(l_{\bar{p}}^c \mathbf{n}_{\bar{p}}^c + l_{\underline{p}}^c \mathbf{n}_{\underline{p}}^c \right) \cdot \mathbf{u}_p = 0, \quad (5a)$$

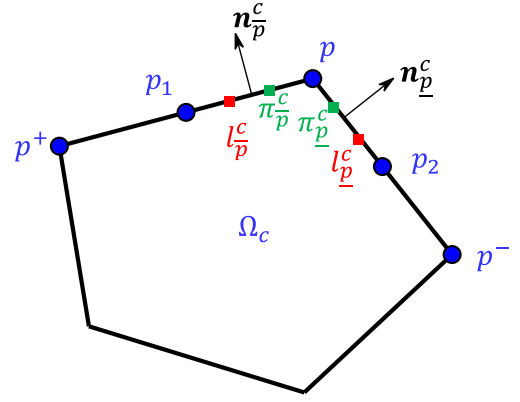


Fig. 1. Notation associated with polygonal cell Ω_c in 2D geometry.

$$m_c \frac{d}{dt} \mathbf{u}_c + \sum_{p \in \mathcal{P}(c)} \left(l_{\bar{p}}^c \pi_{\bar{p}}^c \mathbf{n}_{\bar{p}}^c + l_{\underline{p}}^c \pi_{\underline{p}}^c \mathbf{n}_{\underline{p}}^c \right) = 0, \quad (5b)$$

$$m_c \frac{d}{dt} E_c + \sum_{p \in \mathcal{P}(c)} \left(l_{\bar{p}}^c \pi_{\bar{p}}^c \mathbf{n}_{\bar{p}}^c + l_{\underline{p}}^c \pi_{\underline{p}}^c \mathbf{n}_{\underline{p}}^c \right) \cdot \mathbf{u}_p = 0, \quad (5c)$$

where $\mathbf{u}_p$ is the nodal velocity and m_c , ρ_c , $\mathbf{u}_c$, and E_c are the mass, density, velocity, and total energy of a specific cell c , respectively.

3. Formulation of hybrid nodal solver

3.1. Motivation

A delicate Riemann solver, also known as a nodal solver, is developed in [4]. This nodal solver is genuinely multidimensional because it unites all the cells around each node for information exchange. At each face, the numerical flux is evaluated using standard 1D analysis, which provides a standard procedure. However, the numerical fluxes are evaluated normal to the cell edges, thereby neglecting local wave orientation. An extreme difficulty faced by this method is when waves propagate tangentially to the cell edges. Herein, we investigate the directional effect of wave propagation, beginning with a basic assumption based on the RH jump condition.

3.2. Assumption based on Rankine–Hugoniot condition

In the Lagrangian frame, the 2D Euler equation in differential conservative form is written as

$$\frac{d\mathbf{U}}{dt} + \nabla \cdot \mathbf{F} = 0, \quad (6)$$

with

$$\mathbf{U} = \left(\frac{1}{\rho}, \mathbf{u}, E \right)^T, \quad \mathbf{F} = (-\mathbf{u}, P, P\mathbf{u})^T, \quad (7)$$

where $\mathbf{U}$ is the vector of conservative variables, $\mathbf{F}$ is the flux vector, and ρ , $\mathbf{u}$, E , and P are the density, velocity, total energy, and pressure, respectively. For this system, the RH condition is given as [18]:

$$-s[\![\mathbf{U}]\!] + [\![\mathbf{F}]\!] = 0, \quad (8)$$

where $[\![\cdot]\!]$ denotes the jump through the discontinuity surface, and s is the mass flux swept by the Lagrangian discontinuity. For the momentum equation, the RH condition is given as

$$-s[\![\mathbf{u}]\!] + [\![\mathbf{f}]\!] = 0, \quad (9)$$

where $[\![\mathbf{u}]\!]$ is the velocity jump and $[\![\mathbf{f}]\!]$ is the momentum flux jump. In finite-volume schemes, the dimensional splitting method is often used to obtain the 1D approximation of the above RH condition. Consequently, the velocity jump preserves only the component in the edge normal direction.

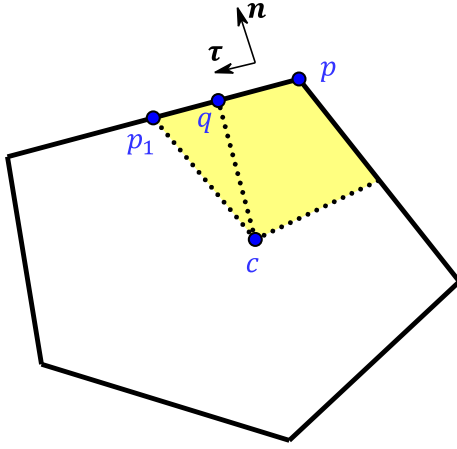


Fig. 2. Sketch of a yellow subcell extracted from Fig. 1.

Herein, instead of the edge normal direction, we assume that waves always propagate along the projection of the velocity gradient to $\mathbf{n}$, which is denoted as $\nabla \mathbf{u} \cdot \mathbf{n}$. From the derivations of discrete operators in [21], it is known that for a given direction $\mathbf{n}$, $\nabla \mathbf{u} \cdot \mathbf{n}$ represents the directional derivative of the velocity field, i.e.,

$$\nabla \mathbf{u} \cdot \mathbf{n} = \frac{\partial \mathbf{u}}{\partial l}. \quad (10)$$

3.3. Discrete form of velocity jump

In this subsection, we construct a discrete form of the velocity jump. Fig. 2 shows a yellow subcell extracted from Fig. 1, where $\mathbf{n}$ and $\boldsymbol{\tau}$ denote the unit outward normal vector and unit tangential vector of half edge pp_1 , respectively. Let cq represent the line perpendicular to edge pp_1 through the intersection point q , and the end points of line cq are c and q . From Eq. (10), the discrete form of $\nabla \mathbf{u} \cdot \mathbf{n}$ is represented as $(\mathbf{u}_q - \mathbf{u}_c) / l_{cq}$, where $\mathbf{u}_q$ and $\mathbf{u}_c$ are the velocities of q and c , respectively, and l_{cq} is the length of line cq . Consequently, the jump relation is built between c and q in discrete space so that the velocity jump is defined as

$$\llbracket \mathbf{u} \rrbracket = \mathbf{u}_q - \mathbf{u}_c = \Delta \mathbf{u}_{cq}. \quad (11)$$

Because q is an auxiliary point in the derivation, the value of $\Delta \mathbf{u}_{cq}$ can be approximated from the primary variables $\mathbf{u}_c$ and $\mathbf{u}_p$.

(1) Using a linear distribution of velocity along the half edge pp_1 , $\Delta \mathbf{u}_{cq}$ is approximated by a dimensional splitting method, i.e.,

$$\llbracket \mathbf{u} \rrbracket_1 = \Delta \mathbf{u}_{cq} \approx (\Delta \mathbf{u}_{cp} \cdot \mathbf{n}) \mathbf{n}, \quad (12)$$

here the subscript 1 is used to represent linear distribution approximation, and $\Delta \mathbf{u}_{cp} = \mathbf{u}_p - \mathbf{u}_c$. The above equation means that waves always propagate along edge normal direction. Substituting Eq. (12) into Eq. (9) gives

$$s_1 (\Delta \mathbf{u}_{cp} \cdot \mathbf{n}) \mathbf{n} = \llbracket \mathbf{f} \rrbracket_1, \quad (13)$$

here the mass flux and numerical flux also use the subscript 1.

(2) Using a constant distribution of velocity along the half edge pp_1 , $\Delta \mathbf{u}_{cq}$ is approximated as

$$\llbracket \mathbf{u} \rrbracket_2 = \Delta \mathbf{u}_{cq} \approx \Delta \mathbf{u}_{cp}, \quad (14)$$

here the subscript 2 is used to represent constant distribution approximation. This equation means that waves always propagate along the direction of $\Delta \mathbf{u}_{cp}$. Substituting Eq. (14) into Eq. (9) gives

$$s_2 \Delta \mathbf{u}_{cp} = \llbracket \mathbf{f} \rrbracket_2, \quad (15)$$

here the mass flux and numerical flux also use the subscript 2.

From above two cases, we deduce that the velocity jump $\llbracket \mathbf{u} \rrbracket$ in Eq. (11) can be defined in different forms, i.e., $(\Delta \mathbf{u}_{cp} \cdot \mathbf{n}) \mathbf{n}$ or $\Delta \mathbf{u}_{cp}$.

3.4. Determination of Lagrangian mass flux

Deriving the Lagrangian mass flux is the main topic in Section 2.3.3 of [22], and here we review its formulation briefly. The jump relation that expresses mass conservation across the discontinuity surface Σ moving with speed $\mathbf{W}$ is written as

$$\llbracket \rho J \rrbracket (\mathbf{W} \cdot \mathbf{n}_\Sigma) = 0, \quad (16)$$

where J is the Jacobian of the fluid map flow with $\rho J = \rho^0$, where ρ^0 is the initial density. Then the mass flux swept by the discontinuity is written as

$$s = \rho_a^0 (\mathbf{W} \cdot \mathbf{n}_\Sigma) = \rho_b^0 (\mathbf{W} \cdot \mathbf{n}_\Sigma), \quad (17)$$

where the subscripts a and b denote the fluid variables across the discontinuity, and the subscript Σ denotes the summation of unit edge normal vectors.

With Eq. (17), we show how the above mass fluxes s_1 and s_2 are evaluated. Assume that a discontinuity moves across the cell interface with a speed $\mathbf{W}$, the initial density of the cell is ρ_c , and the unit normal vector across the interface is $\mathbf{n}$. If the discontinuity moves along the edge normal direction, i.e., $\mathbf{W} = |\mathbf{W}| \mathbf{n}$, then the mass flux s_1 is evaluated as

$$s_1 = \rho_c |\mathbf{W}|. \quad (18)$$

If the discontinuity is in the direction of $\Delta \mathbf{u}_{cp}$, i.e., $\mathbf{W} = \frac{|\mathbf{W}|}{|\Delta \mathbf{u}_{cp}|} \Delta \mathbf{u}_{cp}$, then the mass flux s_2 is evaluated as

$$s_2 = \rho_c \frac{|\Delta \mathbf{u}_{cp} \cdot \mathbf{n}|}{|\Delta \mathbf{u}_{cp}|} |\mathbf{W}| = s_1 \frac{|\Delta \mathbf{u}_{cp} \cdot \mathbf{n}|}{|\Delta \mathbf{u}_{cp}|}, \quad (19)$$

where the velocity difference must satisfy $|\Delta \mathbf{u}_{cp}| \neq 0$. The peculiar condition when $|\Delta \mathbf{u}_{cp}| = 0$ will be discussed later.

3.5. Formulation of hybrid nodal solver

The previous subsections show that with a linear or constant velocity approximation along the half edge, the momentum RH equation can be written in two different forms, i.e., Eq. (13) or (15). More generally, if we use a hybridization of the linear and constant approximations of velocity along the half edge, then $\Delta \mathbf{u}_{cq}$ is approximated as

$$\llbracket \mathbf{u} \rrbracket_h = \Delta \mathbf{u}_{cq} \approx \lambda (\Delta \mathbf{u}_{cp} \cdot \mathbf{n}) \mathbf{n} + (1 - \lambda) \Delta \mathbf{u}_{cp}, \quad (20)$$

here the subscript h is used to represent hybridization, and the coefficient $\lambda \in [0, 1]$. The momentum RH equation is then written as

$$s_h [\lambda (\Delta \mathbf{u}_{cp} \cdot \mathbf{n}) \mathbf{n} + (1 - \lambda) \Delta \mathbf{u}_{cp}] = \llbracket \mathbf{f} \rrbracket_h. \quad (21)$$

Similar to the derivation of Eq. (19), the mass flux s_h is expressed as

$$\begin{aligned} s_h &= s_1 \frac{|\lambda (\Delta \mathbf{u}_{cp} \cdot \mathbf{n}) \mathbf{n} + (1 - \lambda) \Delta \mathbf{u}_{cp} \cdot \mathbf{n}|}{|\lambda (\Delta \mathbf{u}_{cp} \cdot \mathbf{n}) \mathbf{n} + (1 - \lambda) \Delta \mathbf{u}_{cp}|} \\ &= s_1 \frac{|\Delta \mathbf{u}_{cp} \cdot \mathbf{n}|}{|\lambda (\Delta \mathbf{u}_{cp} \cdot \mathbf{n}) \mathbf{n} + (1 - \lambda) \Delta \mathbf{u}_{cp}|}. \end{aligned} \quad (22)$$

Note that the direction of $\llbracket \mathbf{f} \rrbracket_h$ in Eq. (21) changes with λ . However, it can be proved that the absolute value of $\llbracket \mathbf{f} \rrbracket_h$ is always equivalent to $s_1 |\Delta \mathbf{u}_{cp} \cdot \mathbf{n}|$. An explanation for this is that by hybridization, an oblique wave pattern is identified in the 1D configuration. Consequently, the numerical flux is oblique to the interface, but its value does not change. We view Eq. (21) as being just like a rotation of the dimensional splitting RH equation (13) from the $\mathbf{n}$ direction to the $[\lambda (\Delta \mathbf{u}_{cp} \cdot \mathbf{n}) \mathbf{n} + (1 - \lambda) \Delta \mathbf{u}_{cp}]$ direction.

In formulating the nodal solver from Eq. (21), we use the notations in Fig. 1. The superscript c is used to denote the associated cell Ω_c .

Following the approach suggested by Dukowicz [23], the mass flux s_1 is defined as

$$s_1 = \rho_c \left[a_c + \Gamma_c \left| \Delta \mathbf{u}_{cp} \cdot \mathbf{n} \right| \right], \quad (23)$$

and for the half edges pp^+ and pp^- we have

$$s_{p,1}^c = \rho_c \left[a_c + \Gamma_c \left| \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right| \right], \quad (24a)$$

$$s_{p,1}^c = \rho_c \left[a_c + \Gamma_c \left| \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right| \right]. \quad (24b)$$

Hereinafter, the subscript 1 is omitted for simplicity. Then the momentum RH conditions for the half edges pp^+ and pp^- are written as

$$-s_p^c \frac{\left| \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right| \left[\lambda \left(\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right) \mathbf{n}_p^c + (1-\lambda) \Delta \mathbf{u}_{cp} \right]}{\left| \lambda \left(\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right) \mathbf{n}_p^c + (1-\lambda) \Delta \mathbf{u}_{cp} \right|} + \left(P_c - \pi_p^c \right) \mathbf{n}_p^c = 0, \quad (25a)$$

$$-s_p^c \frac{\left| \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right| \left[\lambda \left(\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right) \mathbf{n}_p^c + (1-\lambda) \Delta \mathbf{u}_{cp} \right]}{\left| \lambda \left(\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right) \mathbf{n}_p^c + (1-\lambda) \Delta \mathbf{u}_{cp} \right|} + \left(P_c - \pi_p^c \right) \mathbf{n}_p^c = 0. \quad (25b)$$

For simplicity, we define the following expressions:

$$\alpha_p^c = \frac{\left| \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right|}{\left| \lambda \left(\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right) \mathbf{n}_p^c + (1-\lambda) \Delta \mathbf{u}_{cp} \right|}, \quad (26a)$$

$$\alpha_p^c = \frac{\left| \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right|}{\left| \lambda \left(\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right) \mathbf{n}_p^c + (1-\lambda) \Delta \mathbf{u}_{cp} \right|}. \quad (26b)$$

Eq. (25) is then reformulated as

$$-s_p^c \alpha_p^c \left[\lambda \left(\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right) \mathbf{n}_p^c + (1-\lambda) \Delta \mathbf{u}_{cp} \right] + \left(P_c - \pi_p^c \right) \mathbf{n}_p^c = 0, \quad (27a)$$

$$-s_p^c \alpha_p^c \left[\lambda \left(\Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right) \mathbf{n}_p^c + (1-\lambda) \Delta \mathbf{u}_{cp} \right] + \left(P_c - \pi_p^c \right) \mathbf{n}_p^c = 0, \quad (27b)$$

and the conservation condition around a node is written as

$$\sum_{c \in C(p)} \left(I_p^c \pi_p^c \mathbf{n}_p^c + I_p^c \pi_p^c \mathbf{n}_p^c \right) = 0. \quad (28)$$

Substituting Eq. (27) into Eq. (28) and rearranging, we have

$$M_p \mathbf{u}_p = \sum_{c \in C(p)} \left[I_p^c P_c \mathbf{n}_p^c + I_p^c P_c \mathbf{n}_p^c - M_{cp} \mathbf{u}_c \right], \quad (29)$$

where the 2×2 matrix M_{cp} is defined as

$$M_{cp} = \lambda \left(\alpha_p^c I_p^c \mathbf{n}_p^c \otimes \mathbf{n}_p^c + \alpha_p^c I_p^c \mathbf{n}_p^c \otimes \mathbf{n}_p^c \right) + (1-\lambda) \left(\alpha_p^c I_p^c \mathbf{n}_p^c + \alpha_p^c I_p^c \mathbf{n}_p^c \right) I, \quad (30)$$

I is the identity matrix, and $M_p = \sum_{c \in C(p)} M_{cp}$.

Given $\lambda = 1$, the 2×2 matrix is simplified as

$$M_{cp,1} = s_p^c I_p^c \mathbf{n}_p^c \otimes \mathbf{n}_p^c + s_p^c I_p^c \mathbf{n}_p^c \otimes \mathbf{n}_p^c, \quad (31)$$

and we find the present nodal solver exactly replicates the one developed by Maire [4].

Given $\lambda = 0$, the 2×2 matrix is simplified as

$$M_{cp,2} = \left(\frac{s_p^c I_p^c \left| \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right|}{\left| \Delta \mathbf{u}_{cp} \right|} + \frac{s_p^c I_p^c \left| \Delta \mathbf{u}_{cp} \cdot \mathbf{n}_p^c \right|}{\left| \Delta \mathbf{u}_{cp} \right|} \right) I, \quad (32)$$

and we find that the present nodal solver exactly replicates the one developed by Burton et al. [20]. However, note that although their nodal solver is replicated, their derivations differ from ours. The cornerstone of the nodal solver of Burton et al. [20] is that the stress jump must

be symmetric to satisfy equilibrium of angular momentum, then they develop a tensor dissipation model to satisfy the equilibrium condition. By contrast, the present nodal solver is derived by means of a proper approximation of the velocity jump in the RH condition.

For implementing the present nodal solver, note that M_{cp} in Eq. (30) is not a simple combination of $M_{cp,1}$ and $M_{cp,2}$ with the coefficient λ , and incorporating α_p^c and α_p^c should also be considered. Also, for smooth flow where $\mathbf{u}_p = \mathbf{u}_c$, $M_{cp,2}$ is meaningless; under this peculiar condition, λ should be set to a small number larger than zero, such as 10^{-6} .

4. Second-order scheme

4.1. Spatial extension

The classical least-squares algorithm [4] is applied to reconstruct the fluid variables in each cell, i.e.,

$$W_c(\mathbf{X}_p) = W_c + \nabla W_c \cdot (\mathbf{X}_p - \mathbf{X}_c), \quad (33)$$

where W_c represents the reconstructed pressure or velocity components in cell c , and ∇W_c is the gradient of W around cell c . The gradient is obtained by solving the following minimization problem:

$$\nabla W_c = \operatorname{argmin}_{d \in C(c)} \left[W_d - W_c - \nabla W_c \cdot (\mathbf{X}_d - \mathbf{X}_c) \right]^2, \quad (34)$$

where $C(c)$ is the set of cells neighboring cell c . The solution of this problem can be written as

$$\nabla W_c = M_c^{-1} \sum_{d \in C(c)} (W_d - W_c) (\mathbf{X}_d - \mathbf{X}_c), \quad (35)$$

where M_c is defined as

$$M_c = \sum_{d \in C(c)} (\mathbf{X}_d - \mathbf{X}_c) \otimes (\mathbf{X}_d - \mathbf{X}_c), \quad (36)$$

To preserve monotonicity in the Lagrangian framework with moving meshes, a vector limiter is required [24]. Consider a local coordinate system with basis vectors $\mathbf{e}_1$ and $\mathbf{e}_2$. The transformation matrix $A_p = (\mathbf{e}_p^1, \mathbf{e}_p^2)$ can then be defined from the original coordinate system to the local coordinate system. The vector limiter is

$$\phi_c = A_c^{-1} \begin{pmatrix} \phi_c^1 & 0 \\ 0 & \phi_c^2 \end{pmatrix} A_c, \quad (37)$$

where ϕ_p^1 and ϕ_p^2 are calculated using the Barth–Jespersen method [25], with the velocity field being projected onto the directions of $\mathbf{e}_p^1$ and $\mathbf{e}_p^2$, respectively. In practice, we multiply the limiter by a coefficient $0 \leq \eta \leq 1$ to make it more flexible; η is set to 0.95 if not otherwise specified. For the second-order extension, the following limited reconstruction of the velocity is performed:

$$W_c(\mathbf{X}_p) = W_c + \eta \phi_c \nabla W_c \cdot (\mathbf{X}_p - \mathbf{X}_c). \quad (38)$$

4.2. Time discretization

A two-stage, second-order Runge–Kutta scheme is used to integrate the expressions in Eq. (5). Let $\mathbf{U}$ represent the unknowns ($1/\rho, \mathbf{u}, E$), and let RHS represent the right-hand side of the semi-discrete equations. Then, the governing equations are simplified as

$$m_c \frac{d}{dt} \mathbf{U}_c = \text{RHS}. \quad (39)$$

The time integration of Eqs. (3) and (5) from time n to $n+1$ is thus given as follows:

Stage 1:

$$\mathbf{X}_p^{(1)} = \mathbf{X}_p^n + \Delta t \mathbf{u}_p^n, \quad (40a)$$

$$\mathbf{U}_c^{(1)} = \mathbf{U}_c^n + \Delta t m_c^{-1} \text{RHS}(\mathbf{U}_c^n). \quad (40b)$$

Stage 2:

$$\mathbf{X}_p^{n+1} = \frac{1}{2} \mathbf{X}_p^n + \frac{1}{2} [\mathbf{X}_p^{(1)} + \Delta t \mathbf{u}_p^{(1)}], \quad (41a)$$

$$\mathbf{U}_c^{n+1} = \frac{1}{2} \mathbf{U}_c^n + \frac{1}{2} [\mathbf{U}_c^{(1)} + \Delta t m_c^{-1} \text{RHS}(\mathbf{U}_c^{(1)})]. \quad (41b)$$

The time step is controlled by the CFL condition which is given by

$$\Delta t = C_e \min\{l_c^n / (|\mathbf{u}_c^n| + a_c^n)\}, \quad (42)$$

where the Courant number C_e is set to 0.5, l_c^n is the shortest edge of cell Ω_c , $\mathbf{u}_c^n$ is the velocity of cell Ω_c , and a_c^n is sound speed of cell Ω_c . We also require that the volume change per time step is not larger than ten percent, otherwise the time step will be halved.

4.3. Solution algorithm

In this subsection, we describe the solution algorithm of the numerical scheme in stage 1.

1. The density, pressure, and velocity gradients $\{(\nabla \rho_c)^n, (\nabla P_c)^n, (\nabla \mathbf{u}_c)^n\}$ are computed over a set of cells surrounding Ω_c :

$$(\nabla \rho_c)^n = (M_c^n)^{-1} \sum_{d \in \mathcal{C}(c)} (\rho_d^n - \rho_c^n) (\mathbf{X}_d^n - \mathbf{X}_c^n), \quad (43a)$$

$$(\nabla P_c)^n = (M_c^n)^{-1} \sum_{d \in \mathcal{C}(c)} (P_d^n - P_c^n) (\mathbf{X}_d^n - \mathbf{X}_c^n), \quad (43b)$$

$$(\nabla \mathbf{u}_c)^n = (M_c^n)^{-1} \sum_{d \in \mathcal{C}(c)} (\mathbf{u}_d^n - \mathbf{u}_c^n) (\mathbf{X}_d^n - \mathbf{X}_c^n). \quad (43c)$$

2. The density, pressure, and velocity are reconstructed using the least-squares algorithm:

$$\rho_c(\mathbf{X}_p^n) = \rho_c^n + \eta \phi_c^n (\nabla \rho_c)^n \cdot (\mathbf{X}_p^n - \mathbf{X}_c^n), \quad (44a)$$

$$P_c(\mathbf{X}_p^n) = P_c^n + \eta \phi_c^n (\nabla P_c)^n \cdot (\mathbf{X}_p^n - \mathbf{X}_c^n), \quad (44b)$$

$$\mathbf{u}_c(\mathbf{X}_p^n) = \mathbf{u}_c^n + \eta \phi_c^n (\nabla \mathbf{u}_c)^n \cdot (\mathbf{X}_p^n - \mathbf{X}_c^n). \quad (44c)$$

3. The node velocity is computed iteratively by a fixed-point method:

$$\mathbf{u}_p^n = (M_p^n)^{-1} \sum_{c \in \mathcal{C}(p)} [P_c(\mathbf{X}_p^n) (I_p^{c,n} \pi_p^{c,n} \mathbf{n}_p^{c,n} + I_p^{c,n} \pi_p^{c,n} \mathbf{n}_p^{c,n}) - M_{cp}^n \mathbf{u}_c(\mathbf{X}_p^n)], \quad (45)$$

where the superscript n indicates that the variables are defined at time t^n . The computation of M_{cp}^n involves $\rho_c(\mathbf{X}_p^n)$ and $\mathbf{u}_c(\mathbf{X}_p^n)$, and it is found that the solution converges after only a few iterations.

4. The products of node pressures and edge lengths and normals are presented with Eqs. (25) and (30):

$$I_p^{c,n} \pi_p^{c,n} \mathbf{n}_p^{c,n} = I_p^{c,n} P_c(\mathbf{X}_p^n) \mathbf{n}_p^{c,n} - \alpha_p^{c,n} \Delta \mathbf{u}_a^n, \quad (46a)$$

$$I_p^{c,n} \pi_p^{c,n} \mathbf{n}_p^{c,n} = I_p^{c,n} P_c(\mathbf{X}_p^n) \mathbf{n}_p^{c,n} - \alpha_p^{c,n} \Delta \mathbf{u}_a^n, \quad (46b)$$

where $\Delta \mathbf{u}_a^n$ and $\Delta \mathbf{u}_a^n$ are defined as

$$\Delta \mathbf{u}_a^n = [\lambda (\Delta \mathbf{u}_{cp}^n \cdot \mathbf{n}_p^{c,n}) \mathbf{n}_p^{c,n} + (1 - \lambda) \Delta \mathbf{u}_{cp}^n], \quad (47a)$$

$$\Delta \mathbf{u}_a^n = [\lambda (\Delta \mathbf{u}_{cp}^n \cdot \mathbf{n}_p^{c,n}) \mathbf{n}_p^{c,n} + (1 - \lambda) \Delta \mathbf{u}_{cp}^n], \quad (47b)$$

and $\Delta \mathbf{u}_{cp}^n$ is equal to $\mathbf{u}_p^n - \mathbf{u}_c(\mathbf{X}_p^n)$.

5. The trajectory equation and the governing equations are updated simultaneously:

$$\mathbf{X}_p^{(1)} = \mathbf{X}_p^n + \Delta t \mathbf{u}_p^n, \quad (48a)$$

$$\frac{1}{\rho_c^{(1)}} = \frac{1}{\rho_c^n} + \frac{\Delta t}{m_c} \sum_{p \in \mathcal{P}(c)} (I_p^{c,n} \pi_p^{c,n} \mathbf{n}_p^{c,n} + I_p^{c,n} \pi_p^{c,n} \mathbf{n}_p^{c,n}) \cdot \mathbf{u}_p^n, \quad (48b)$$

$$\mathbf{u}_c^{(1)} = \mathbf{u}_c^n - \frac{\Delta t}{m_c} \sum_{p \in \mathcal{P}(c)} (I_p^{c,n} \pi_p^{c,n} \mathbf{n}_p^{c,n} + I_p^{c,n} \pi_p^{c,n} \mathbf{n}_p^{c,n}), \quad (48c)$$

$$E_c^{(1)} = E_c^n - \frac{\Delta t}{m_c} \sum_{p \in \mathcal{P}(c)} (I_p^{c,n} \pi_p^{c,n} \mathbf{n}_p^{c,n} + I_p^{c,n} \pi_p^{c,n} \mathbf{n}_p^{c,n}) \cdot \mathbf{u}_p^n. \quad (48d)$$

Similarly, the solution algorithm of stage 2 is implemented by Eq. (41). At the end of each stage, the equation of state is called to update the cell pressures.

5. Effectiveness of hybrid nodal solver

In this section, we report several numerical experiments to assess the performance of the hybrid nodal solver with polar grids. Its effectiveness is shown for different values of λ .

1. The Sedov problem [26] describes a diverging shock wave. The computational domain is a quarter circle defined in the region $[0, 1.2] \times [0, 1.2]$. The initial condition of gas with $\gamma = 1.4$ is $[\rho_0, P_0, \mathbf{u}_0] = [1, 10^{-6}, 0]$. This problem is simulated with a 100×20 uniform polar grid.

2. The Richtmyer–Meshkov (RM) instability problem is considered in a piston-driven flow. Initially, two ideal gases are separated by a perturbed interface located at $x = 0$. The left and right states separated by the interface are characterized by $[\rho_l, P_l, \gamma_l] = [1, 1, 1.5]$ and $[\rho_r, P_r, \gamma_r] = [2, 1, 3]$, respectively. The piston moves from the left boundary toward the right boundary in a $[-5, 4.2] \times [0, 0.5]$ box, which is divided into a 460×24 uniform rectangular grid. The piston velocity is 0.603, and the other boundaries are reflecting walls. A small perturbation of the interface is described by

$$x_{\text{pert}} = a_0 \cos(2\pi y), \quad y \in [0, \frac{1}{2}]. \quad (49)$$

where a_0 is set to 10^{-4} , and $\alpha(t)$ is the perturbation amplitude

$$\alpha(t) = \frac{|\Delta x(t)|}{a_0}, \quad (50)$$

where $|\Delta x(t)|$ is the maximum difference of the point abscissa on the perturbed interface.

3. The multi-mode implosion problem [9] with Rayleigh–Taylor (RT) instability is studied in this case. Initially, a cylinder of light fluid ($R \in [0, 1]$) is surrounded by a shell of heavy fluid ($R \in [1, 1.2]$); see Fig. 3. Both fluids have a polytropic index of 5/3. The initial conditions of the light and heavy fluids are $[\rho_l, P_l, \mathbf{u}_l] = [0.05, 0.1, 0]$ and $[\rho_h, P_h, \mathbf{u}_h] = [1, 0.1, 0]$, respectively. The implosion is driven by a source of pressure imposed on the outer surface of the heavy fluid:

$$p^*(t) = \begin{cases} 13 & \text{if } t \in [0, 0.04] \\ 13 - 12.5 \times \frac{t - 0.04}{0.125 - 0.04} & \text{if } t \in [0.04, 0.125] \\ 0.5 & \text{if } t \in [0.125, 0.3]. \end{cases}$$

As can be seen from Fig. 3, a 90×40 polar grid is applied in the quarter-circle domain to simulate this problem. In the angular direction, both fluids have a grid with 40 equal-sized elements. In the radial direction, the light fluid uses a grid with 50 equal-sized elements, and the heavy fluid uses a geometrically graded grid to match the mass at the material interface. The perturbation of the initial interface between the two fluids is

$$r_i^{\text{pert}} = r_i [1 + a_0 \cos(n\theta)],$$

where r_i and r_i^{pert} are the unperturbed and perturbed radii, and a_0 and n are the amplitude and mode of the perturbation. Here a_0 and n are set to 10^{-3} and 8.

5.1. Results for first-order scheme

In this subsection, we report the simulations conducted with the first-order scheme.

Fig. 4 shows the simulation results for the Sedov problem. As λ changes, the simulation results are basically consistent with the

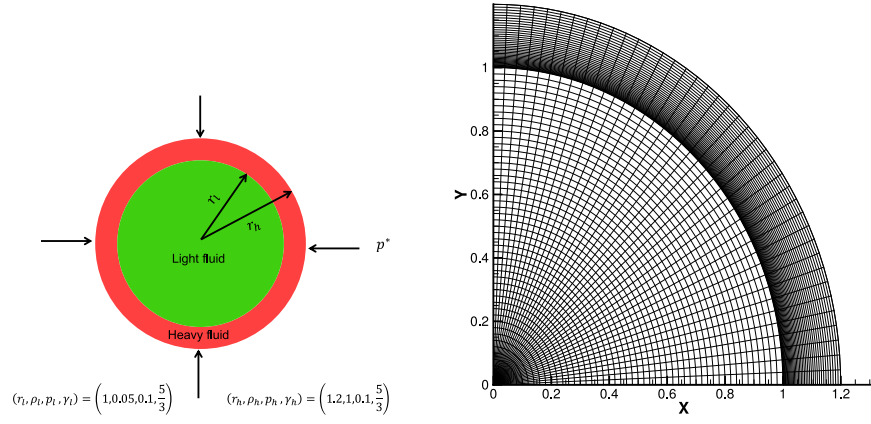


Fig. 3. Initial conditions of the multimode implosion problem. Left: Geometry data; Right: Mesh.

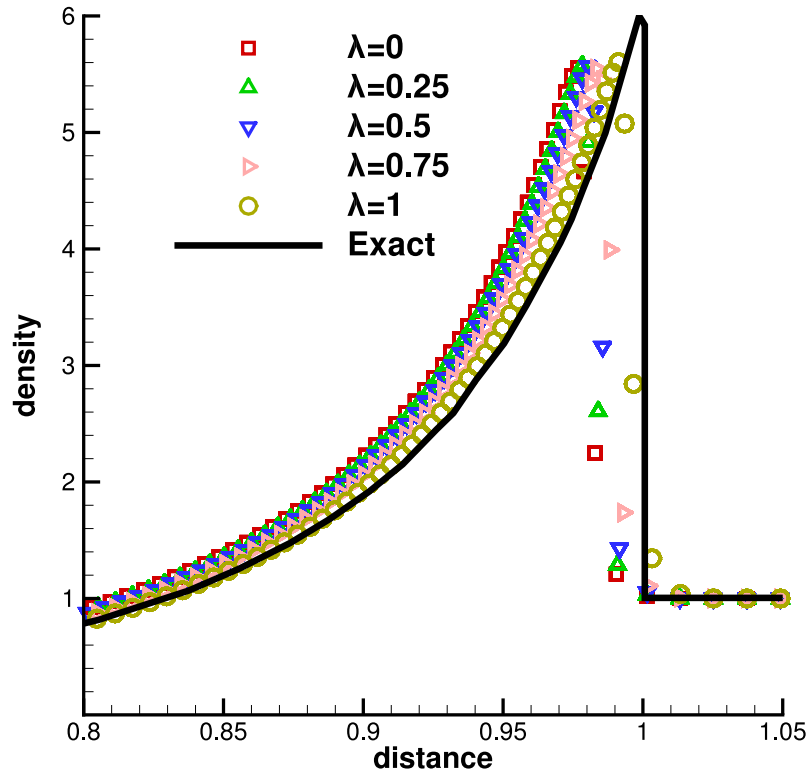


Fig. 4. Density-distance profile for Sedov problem with different values of λ (first-order scheme).

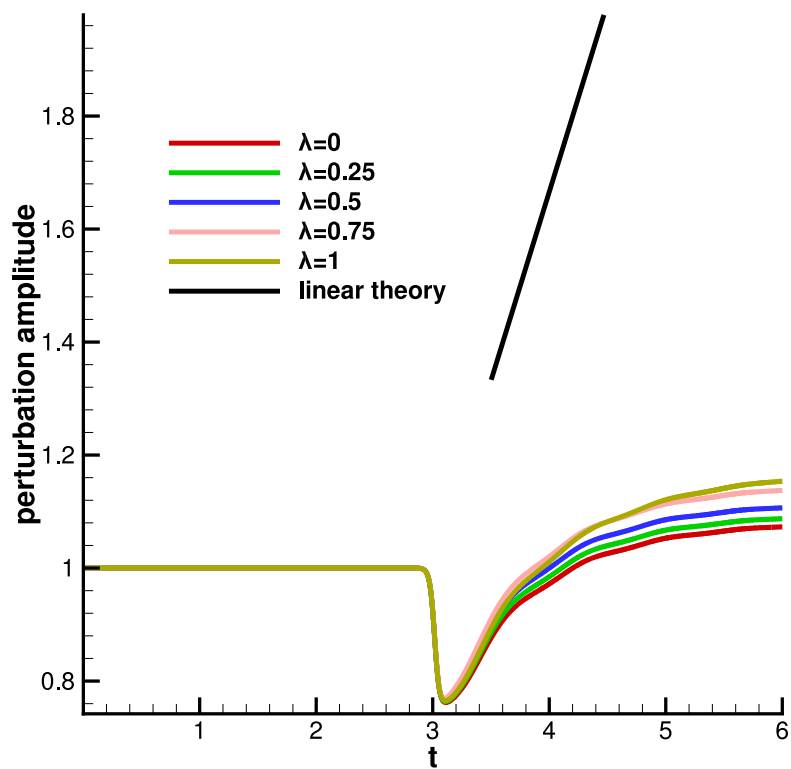


Fig. 5. Richtmyer–Meshkov (RM) instability problem: numerical perturbation amplitude with first-order scheme and comparison with linear theory.

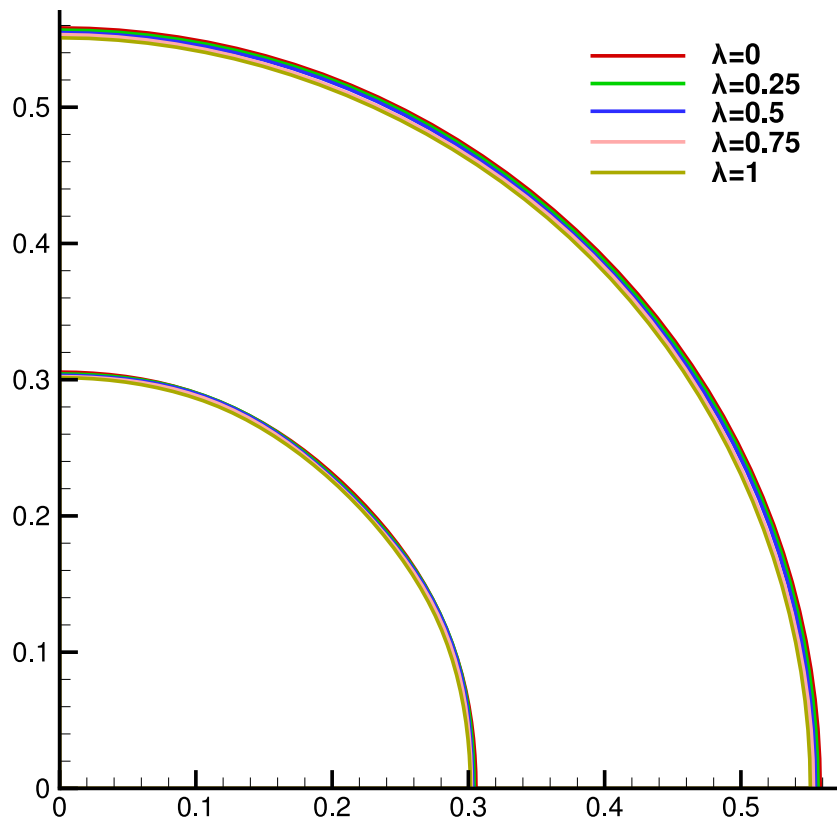


Fig. 6. Multi-mode implosion problem: inner and outer material interfaces at $t = 0.3$ with first-order scheme.

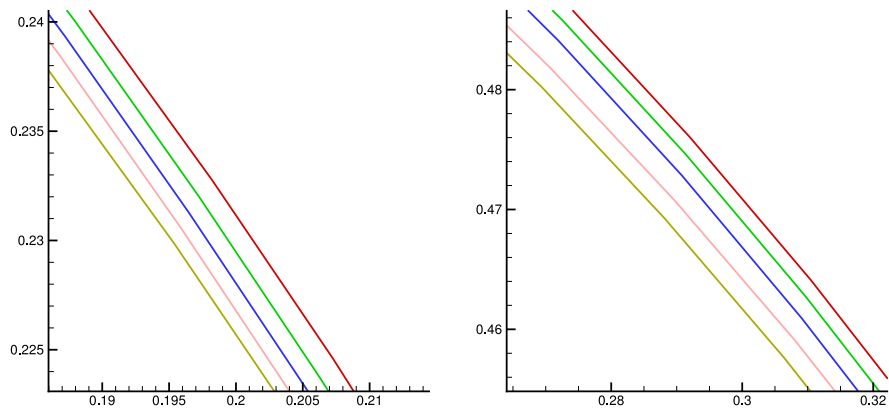


Fig. 7. Enlargement of Fig. 6: inner interface (left); outer interface (right).

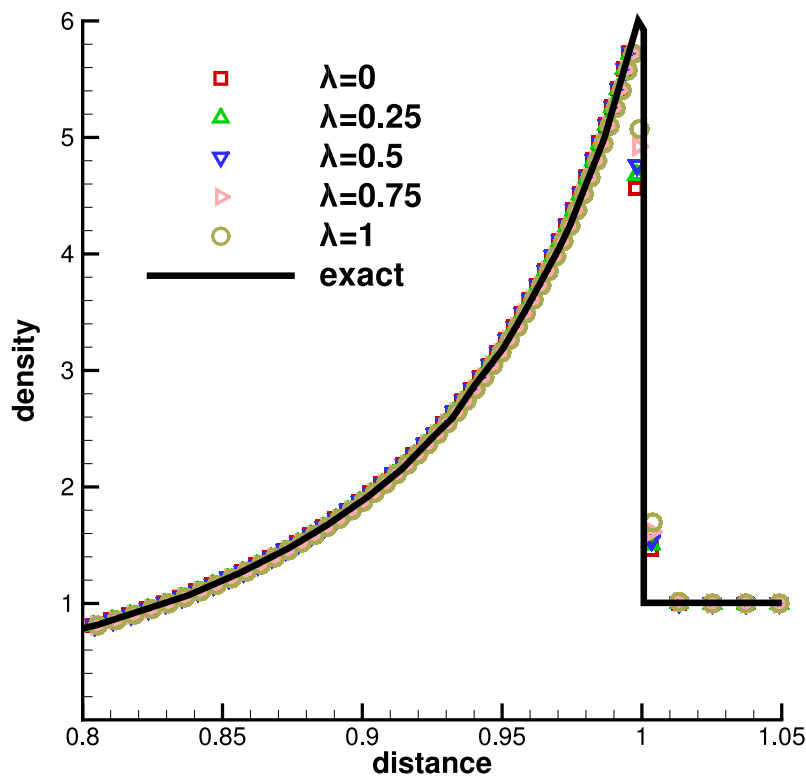


Fig. 8. Density-distance profile for the Sedov problem with different values of λ (second-order scheme).

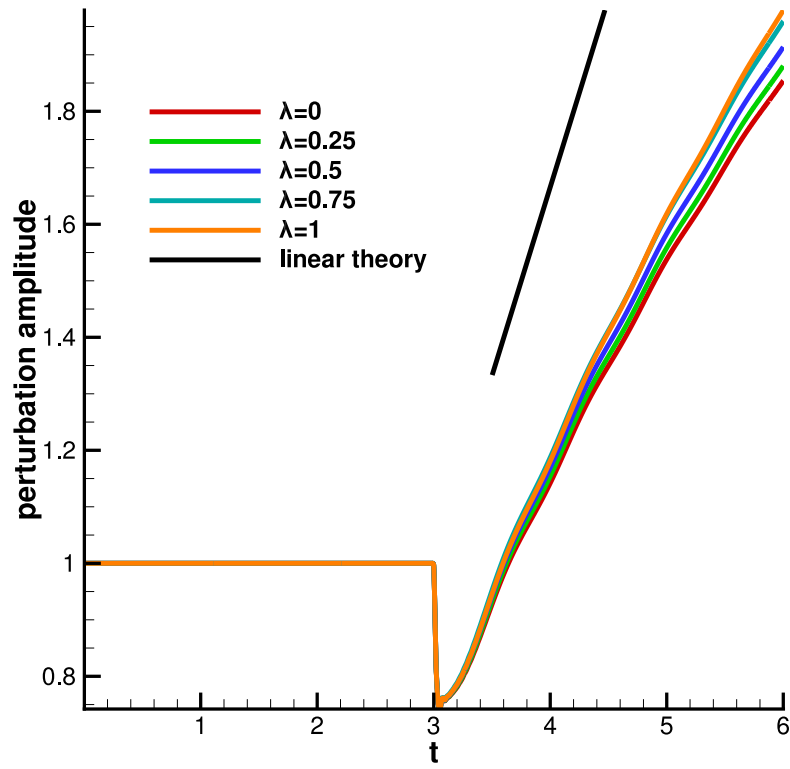


Fig. 9. RM instability problem: numerical perturbation amplitude with second-order scheme and comparison with linear theory.

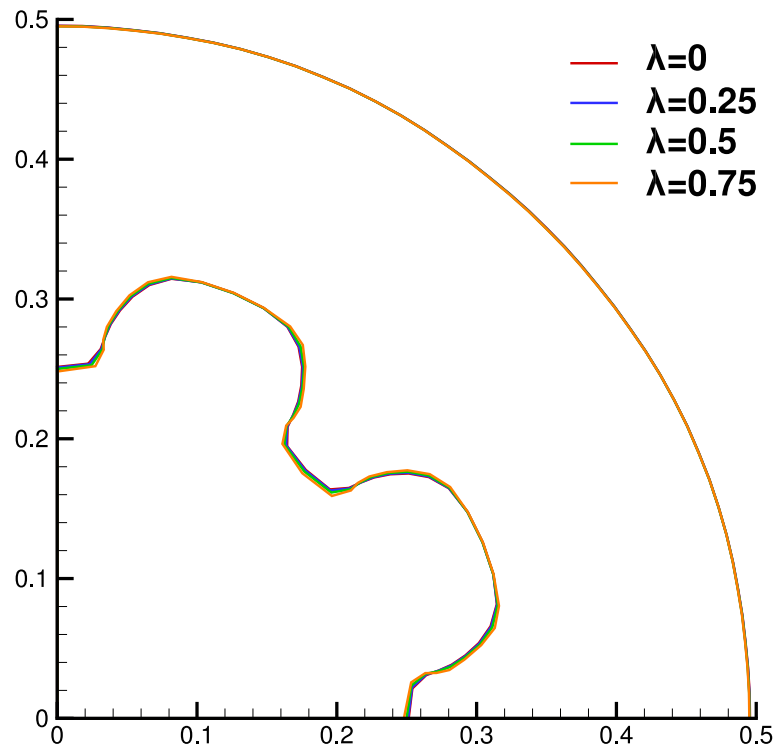


Fig. 10. Multi-mode implosion problem: inner and outer material interfaces at $t = 0.3$ with second-order scheme.

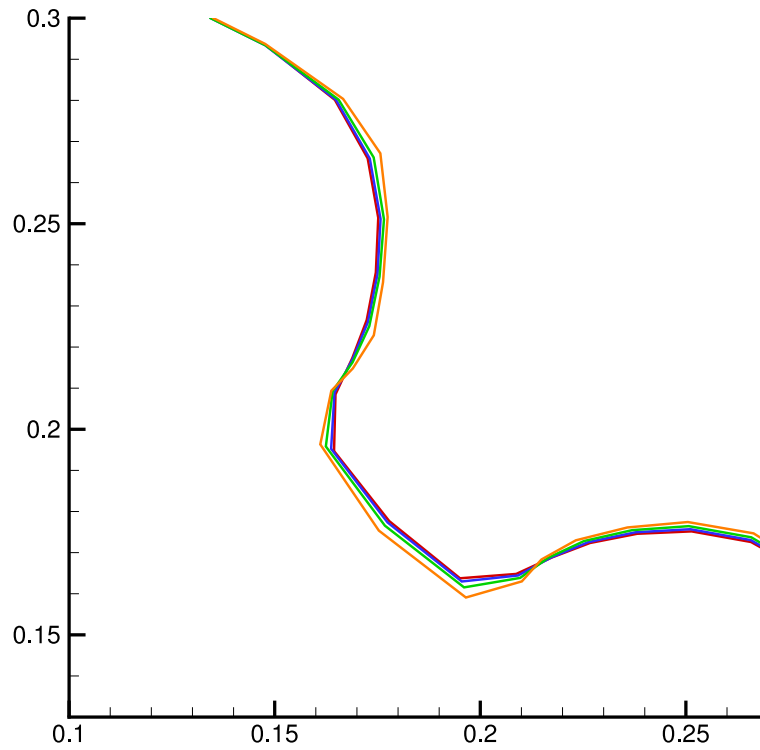


Fig. 11. Enlargement of Fig. 10.

analytical solution. Also, larger λ gives better agreement with the analytical solution, and in our opinion, this is due to the strategy of hybridizing the linear and constant velocity approximations along the mesh half edge [see Eq. (20)]. Basically, larger λ means a higher-order approximation of the velocity variable and hence lower diffusivity, and it is natural that lower diffusivity provides more-accurate results.

For the RM instability problem, Fig. 5 shows the numerical perturbation-amplitude-time profile. As can be seen, the developments of the perturbation amplitude for different values of λ are close to the linear theory. Even more interestingly, as λ is increased, the perturbation amplitude agrees more with the linear theory, and its behavior is similar to that analyzed in the Sedov problem, when λ is increased, the diffusivity decreases and the results become more accurate.

For the multi-mode implosion problem, Fig. 6 shows the material interfaces at $t = 0.3$. Apparently, unlike the arbitrary Lagrangian-Eulerian (ALE) method in [9], here there are no obvious perturbed interfaces. Because the first-order scheme is adopted here, any perturbation is likely to be damped by the high dissipation. Furthermore, Fig. 7 shows enlargements of the inner and outer interfaces. As λ increases, the inner and outer interface radii achieve smaller values; this is because with larger λ , less kinetic energy is dissipated, and so the interface radii achieve smaller values.

To conclude, for the first-order scheme, simulations of the Sedov, RM, and RT instability problems show the effectiveness of the hybrid nodal solver, and that low λ is associated with high dissipation.

5.2. Results for second-order scheme

In this subsection, we report simulations conducted with the second-order scheme. Fig. 8 shows the results for the Sedov problem, which agree well with the analytical solution. Also, the results differ little for different values of λ . Compared to the simulation results for the first-order scheme (see Fig. 4), λ now has less effect on the results.

Fig. 9 shows the results for the RM instability problem. As can be seen, the developments of the perturbation amplitude for different

values of λ are close to that of linear theory. Besides, as λ is increased, the perturbation amplitude agrees more with that of linear theory.

Fig. 10 shows the material interfaces in the multi-mode implosion problem at $t = 0.3$. Clearly, the second-order scheme captures the perturbed interfaces in a similar way as does the ALE method presented in [9]. The enlargement of the inner interface in Fig. 11 shows that smaller λ produces a smoother material interface. This result is expected because smaller λ results in more dissipation to smooth the mesh motions.

Furthermore, Figs. 12 and 13 show the mesh and vorticity field at $t = 0.3$ for different values of λ . As can be seen, mesh distortions occur with high vorticity near the inner material interface; this may be a physical effect due to the presence of shear. Note that in Fig. 12 the mesh distortion worsens with larger λ , and a special case is shown in Fig. 14 for $\lambda = 1$, where the mesh distortion accompanied by high vorticity causes the run to terminate at $t = 0.282$. It is suggested that a potential cause of the mesh distortion is shear. Suppose that a pure shear flow exists; this means that the force resulting from the RH condition should act only in the velocity direction. Thus, the value of λ must be zero, otherwise the force would have a component perpendicular to the velocity direction, which is unphysical. This case shows the potential obstacle of mesh distortion caused by large λ , when the shear effect is present.

To conclude, in simulations of the Sedov, RM, and RT instability problems with the second-order scheme, the results are similar for different λ . This confirms the effectiveness of the hybrid nodal solver. Also, there are small differences among the results for different λ . One point to note is that for flow with strong shear, small λ should be considered to avoid potential mesh distortions.

6. Adaptive hybridization strategy

The numerical experiments showed that large λ offers high accuracy, whereas small λ offers mesh robustness for flows with shear. Therefore, because shear is accompanied by high vorticity, the value

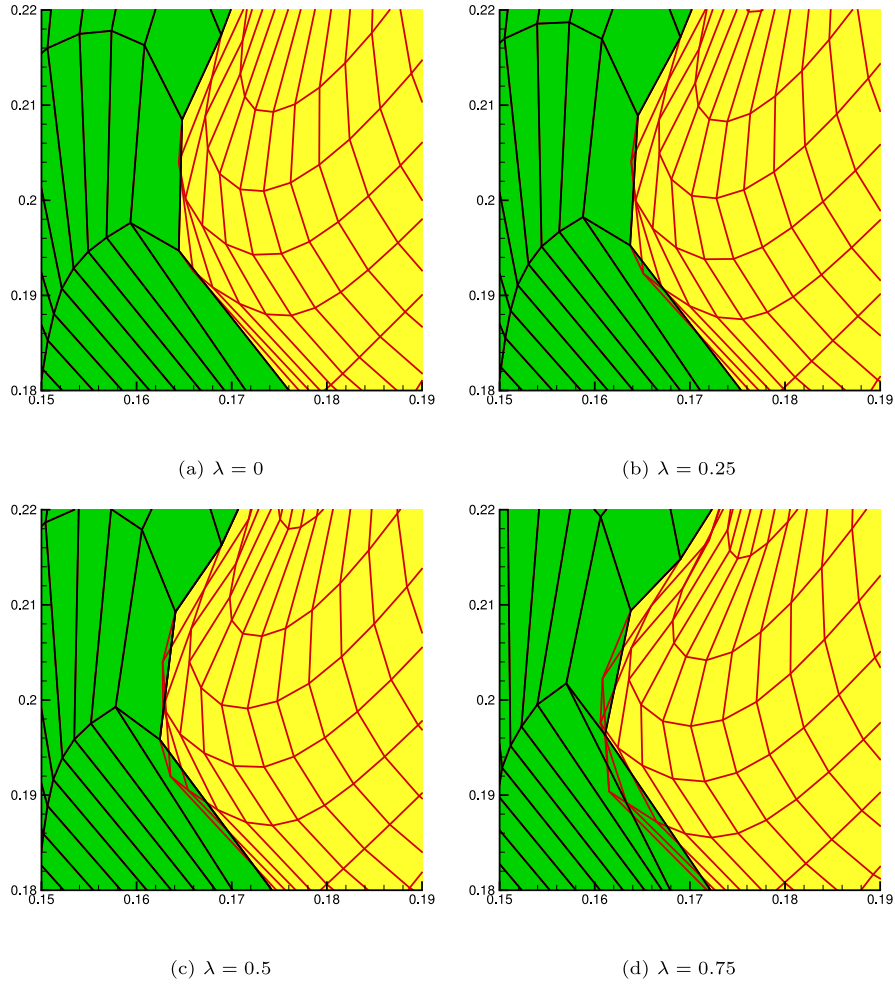


Fig. 12. Mesh near inner material interface for multi-mode implosion problem at $t = 0.3$.

of λ could be chosen adaptively according to the vorticity of the flow field. The vorticity ω is defined as

$$\omega = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}, \quad (51)$$

and the values of $\frac{\partial v}{\partial x}$ and $\frac{\partial u}{\partial y}$ can be obtained from the evaluation of the velocity gradient as described in Section 4.3.

The primary consideration in the adaptive hybridization strategy is that when $|\omega|$ is small, λ should approach 1 to obtain high accuracy, and when $|\omega|$ is large, λ should approach 0 to avoid potentially unphysical results in the case of strong shear. To achieve this, we propose the following empirical function for hybridization:

$$\lambda = e^{-|\omega|/5}, \quad (52)$$

which ensures that λ is close to 1 in cases of low vorticity and approaches 0 rapidly in cases of high vorticity.

7. Simulations with adaptive hybridization

Here, we reinvestigate the Sedov problem in Section 5, and Fig. 15 shows the results of the first-order scheme using adaptive method. As can be seen, the density–distance profile with the adaptive method is more consistent with the results when $\lambda = 1$. This proves the capability of the adaptive method to obtain high accuracy. Furthermore, Fig. 16 shows the results of the second-order scheme using adaptive method. As can be seen, the density–distance profile with the adaptive method is similar to the other cases, and all cases are accurate.

Next, we reinvestigate the RM instability problem in Section 5, and Figs. 17 and 18 show the results of the first-order and second-order schemes using adaptive method. Both figures show that the adaptive method are more consistent with the results when $\lambda = 1$. This implies that the adaptive method achieves high accuracy in both the first-order and second-order schemes.

Next, we reinvestigate the multi-mode implosion problem in Section 5 with the second-order scheme, and Fig. 19 shows the material interfaces at $t = 0.3$. The enlargement of the inner interface shows that the sharpness of the interface with the adaptive method is close to that for $\lambda = 0.25$ and 0.5 . For more details of the adaptation, Fig. 20 shows the evolution of λ .

Furthermore, Fig. 21 shows the mesh near the material interface. The mesh in Fig. 21(f) with the adaptive method, although not stable as Fig. 21(a)–(b), is shown to be less distorted when compared to Fig. 21(c)–(e). This confirms the capability of the adaptive method to enhance mesh robustness.

We also study the adaptive method in cases with Cartesian grids. The Sedov, Saltzman, and Noh problems are considered. To save space, these cases are only simulated using the second-order scheme.

Fig. 22 shows the results of the Sedov problem with a uniform 50×50 initial Cartesian grid. As can be seen, the final grid shows a good circular configuration, and the density–distance profile agrees well with the analytical solution.

The Saltzman problem is a 1D flow problem on a nonuniform 2D grid, and it is often used to test the robustness of Lagrangian schemes. In this problem, a piston moves from the left boundary toward the right in a $[0, 1] \times [0, 0.1]$ box, and the velocity of the piston is 1. The other

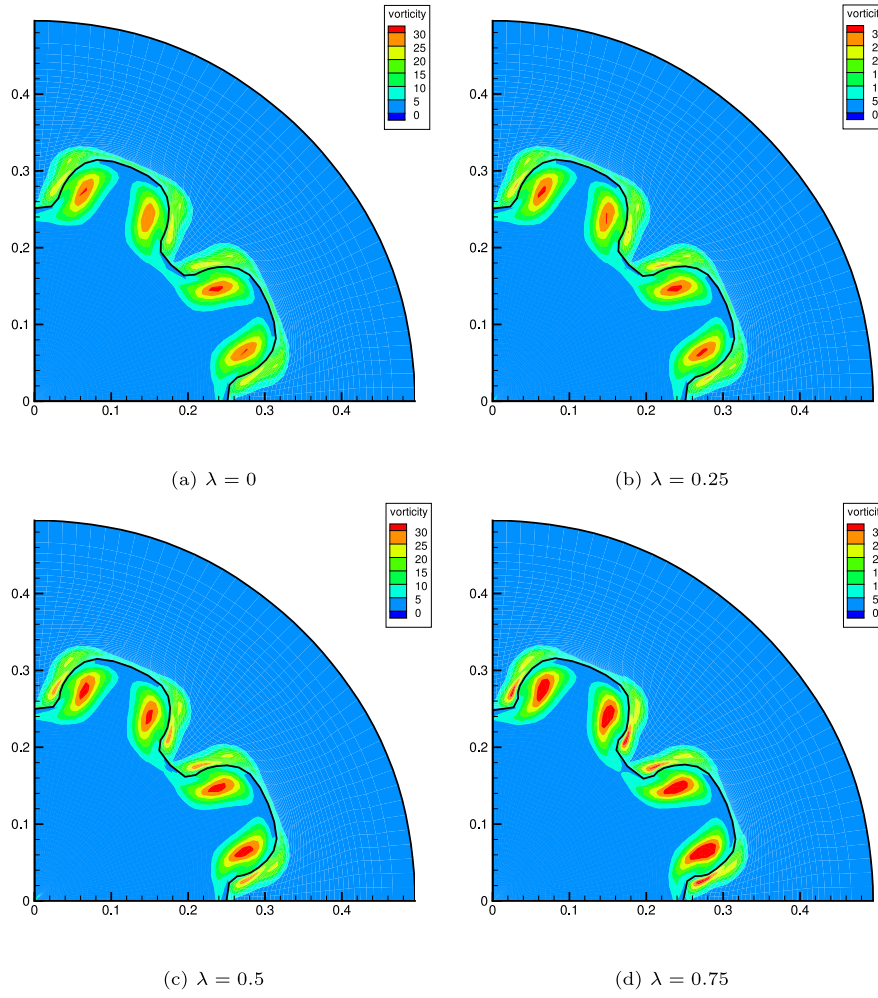


Fig. 13. Vorticity field of multi-mode implosion problem at $t = 0.3$.

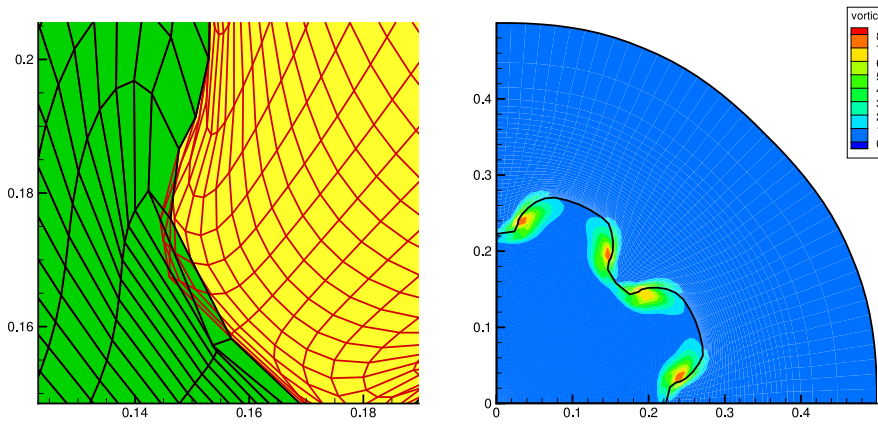


Fig. 14. Mesh and vorticity field near material interface with $\lambda = 1$ at $t = 0.282$.

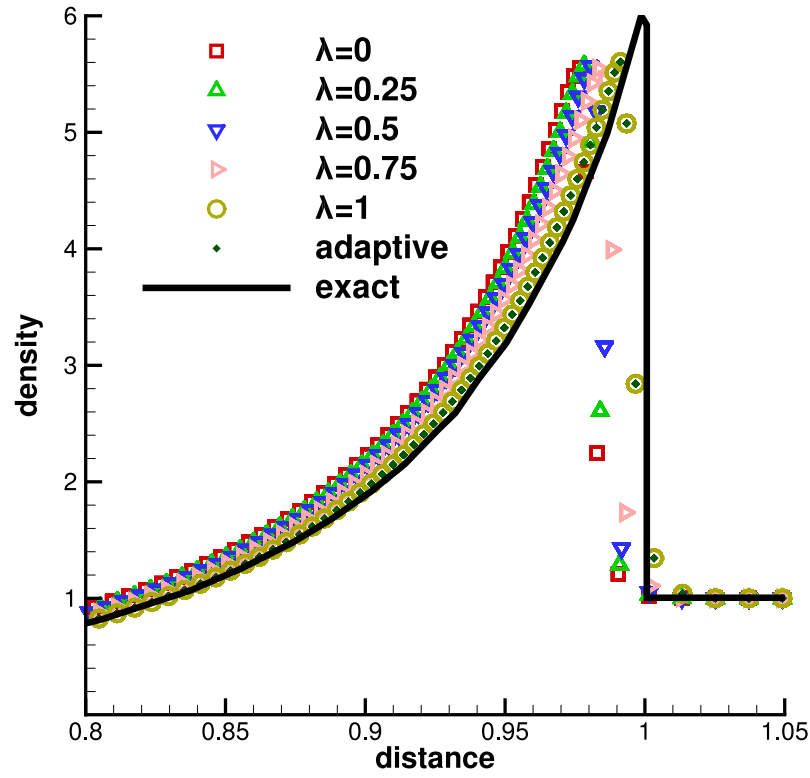


Fig. 15. Density–distance profiles for Sedov problem with different values of λ (first-order scheme with adaptive method).

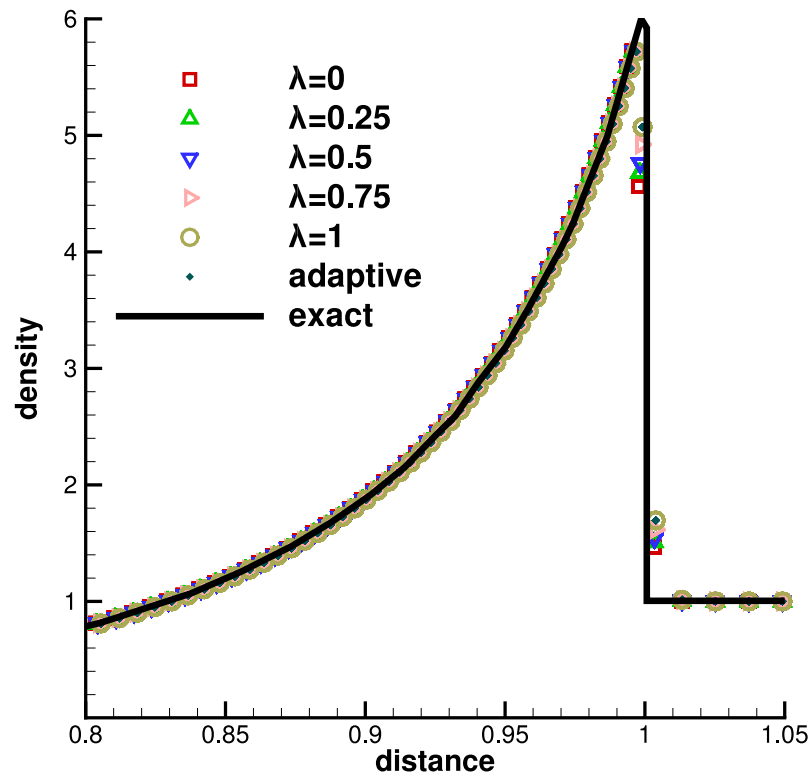


Fig. 16. Density–distance profiles for Sedov problem with different values of λ (second-order scheme with adaptive method).

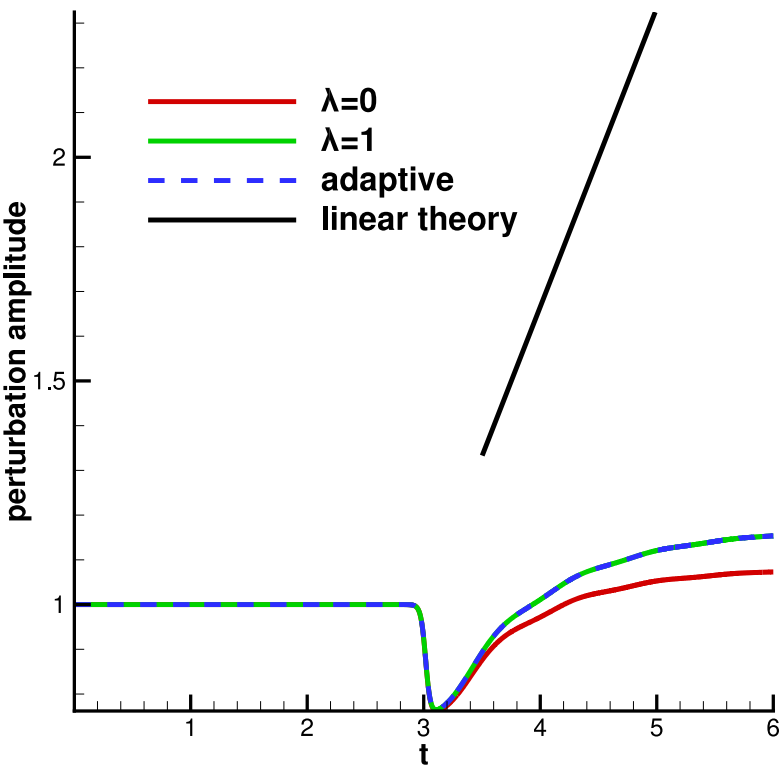


Fig. 17. Numerical perturbation amplitude with first-order scheme using adaptive method.

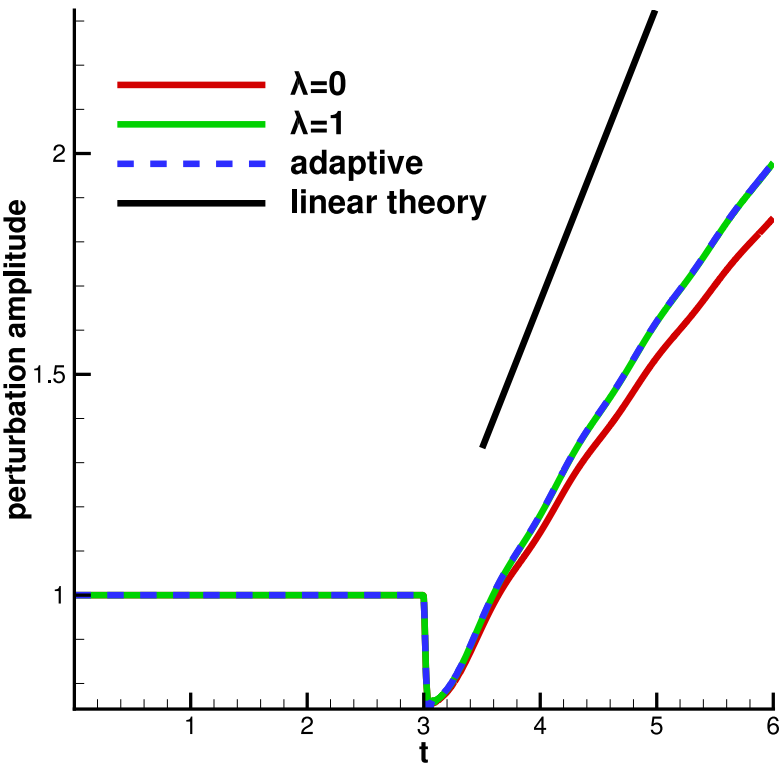


Fig. 18. Numerical perturbation amplitude with second-order scheme using adaptive method.

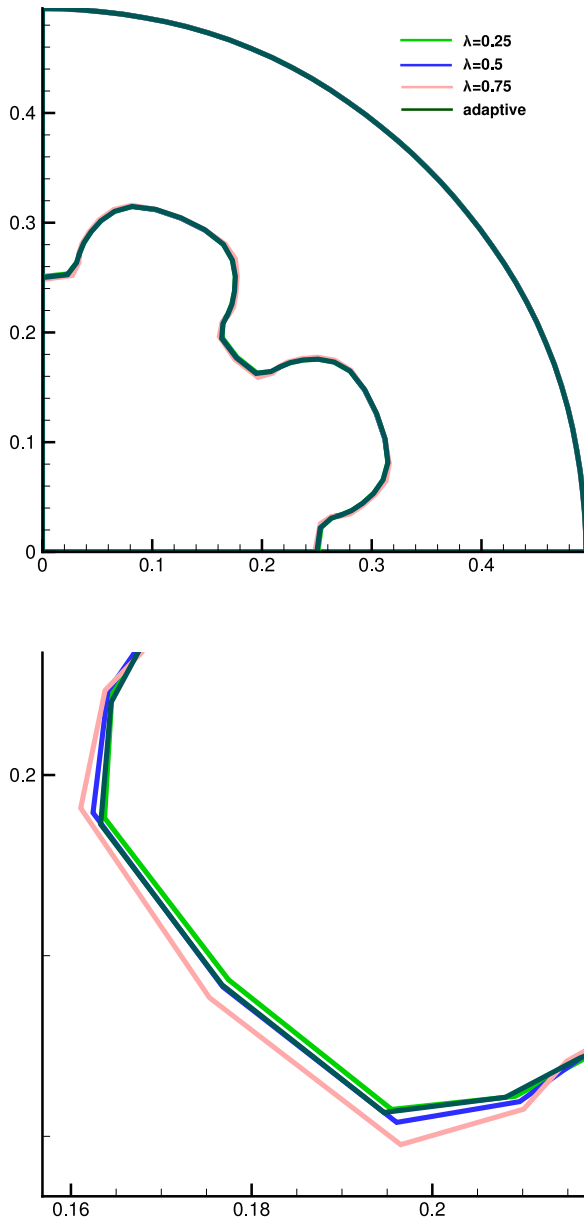


Fig. 19. Material interfaces at $t = 0.3$: global (top); local enlargement of inner interface (bottom).

boundaries are reflecting walls. The distribution of the nonuniform grid is

$$x = (i - 1)\Delta x + (11 - j)\Delta y \sin[\Delta x(i - 1)\pi], \quad y = (j - 1)\Delta y,$$

where $\Delta x = \Delta y = 0.01$. The fluid has a polytropic index of $5/3$. In this problem, the shock wave reflects iteratively between the left and right boundaries. The first shock moves from the left to the right with a constant speed of $4/3$. After reaching the right boundary, the reflected shock moves from the right to the left with a constant speed of $2/3$. Generally, Lagrangian schemes might encounter robustness problems for this case, with the simulations unable to run past $t = 0.8$. Fig. 23 shows the simulation results for this problem at $t = 0.9$ with the adaptive method. The density profile is consistent with the analytical solution, and the mesh preserves a good 1D configuration.

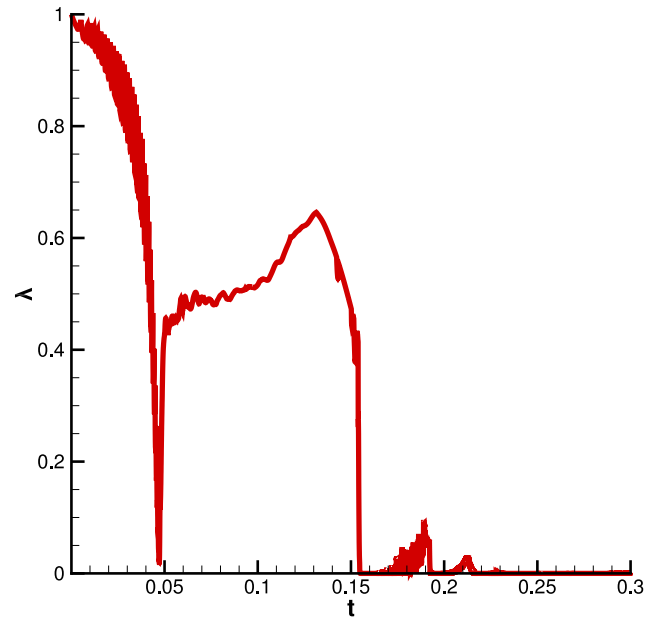


Fig. 20. Evolution of λ with time.

The Noh problem is a well-known test with a strong discontinuity. The computational domain is a quarter circle defined in the region $[0, 1] \times [0, 1]$, the initial conditions are $[\rho_0, \epsilon_0, \mathbf{U}_0(x, y)] = [1, 10^{-6}, (-x/\sqrt{x^2 + y^2}, -y/\sqrt{x^2 + y^2})]$, and the polytropic index γ of the fluid is $5/3$. Initially, a shock wave reflects from the origin with speed $1/3$. For this problem, the analytical solution at radius r and time t is

$$\{\rho, \epsilon, u_r\} = \begin{cases} \left\{ \left(\frac{\gamma + 1}{\gamma - 1} \right)^d, 0.5, 0 \right\} & \text{if } r < r_s, \\ \left\{ \left(1 - \frac{t}{r} \right)^d, 0, 1 \right\} & \text{if } r > r_s, \end{cases}$$

where u_r is the velocity at radius r , $r_s = t/3$ is the position of the shock wave, and $d = 1, 2$, or 3 for 1D, 2D, or 3D problems, respectively.

This problem was implemented on a 50×50 uniform Cartesian grid, and Fig. 24 shows the simulation results at $t = 0.6$. As can be seen, both the mesh distribution and density-distance profile agree well with the analytical solution, and the results verify the effectiveness of the adaptive method.

8. Conclusion

Presented herein was the development of a hybrid nodal solver in the Lagrangian frame. First, from a basic assumption, the velocity jump in the RH condition was used to consider the directional effect of wave propagation. Then by using a combination of constant and linear velocity approximations, an explicit expression for the velocity jump was obtained, resulting in the formulation of a hybrid nodal solver that interestingly replicates two classical ones. Some numerical experiments were reported to show the effectiveness of the hybrid nodal solver. Furthermore, by introducing an adaptive hybridization strategy, the hybrid nodal solver was shown to have the advantages to obtain high accuracy and mesh robustness. The adaptive method also showed good performance on different grid types.

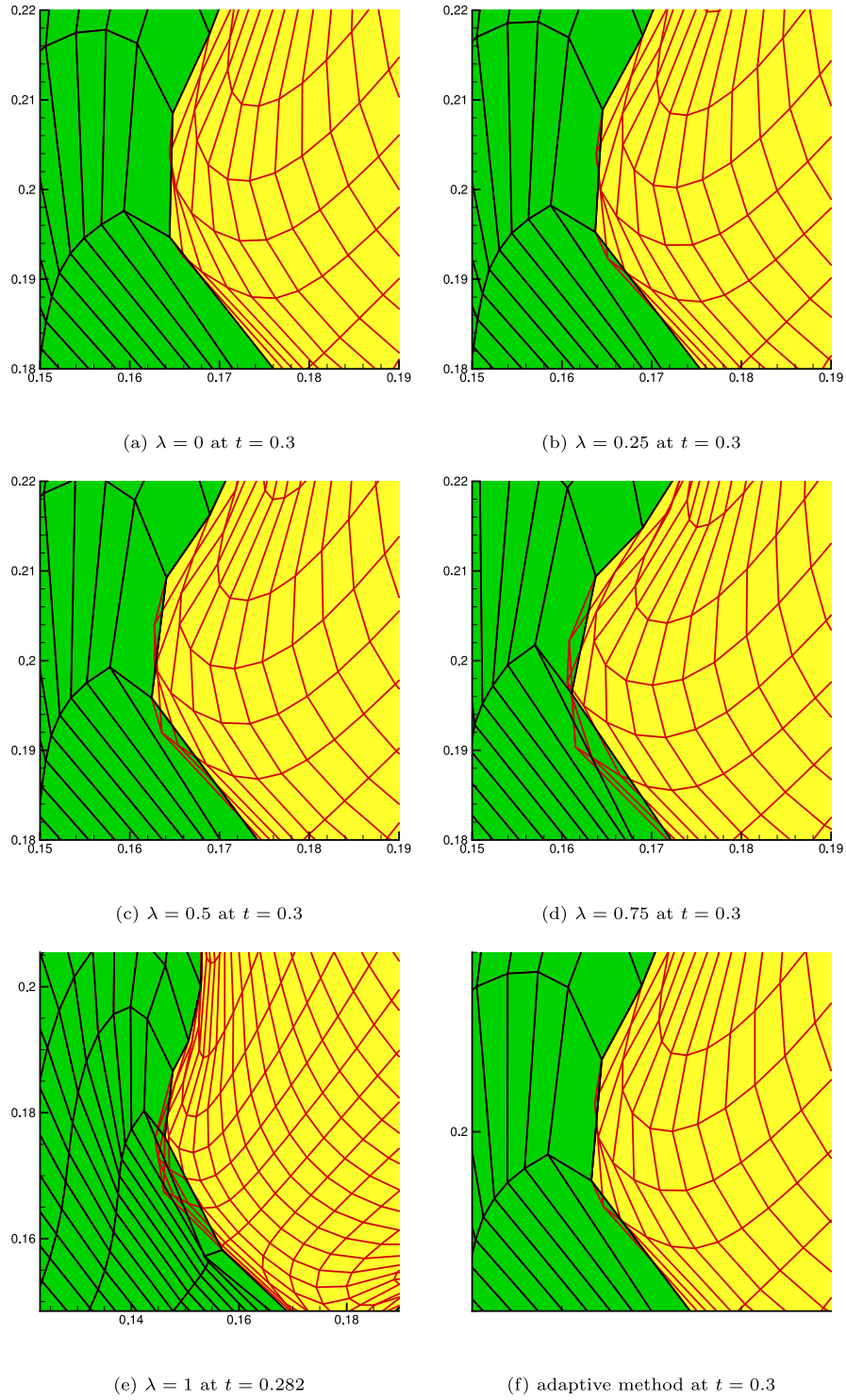


Fig. 21. Mesh near material interface with and without adaptive method.

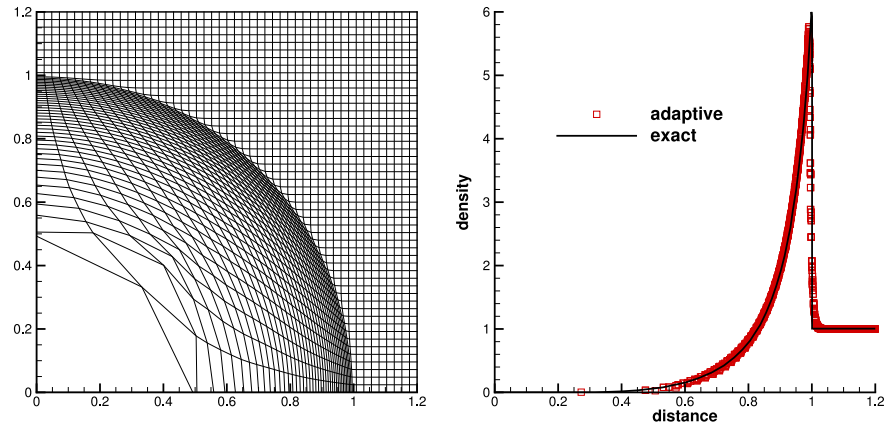


Fig. 22. Simulation results for Sedov problem at $t = 1$ with 50×50 initial Cartesian grid: final mesh (left); density–distance profile with adaptive method (right).

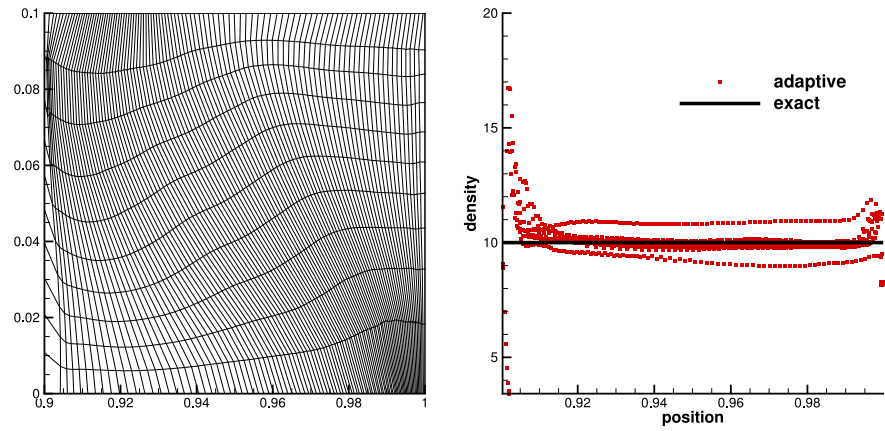


Fig. 23. Simulation results for Saltzman problem at $t = 0.9$ with 100×10 initial Cartesian grid: final mesh (left); density–distance profile with adaptive method (right).

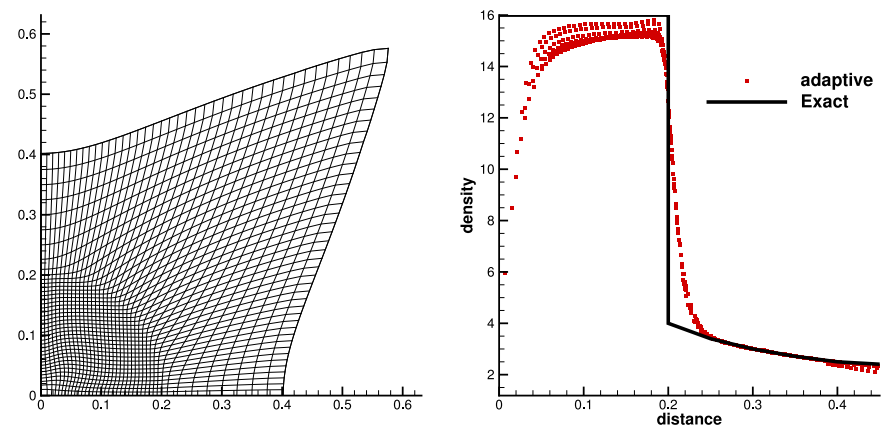


Fig. 24. Simulation results for Noh problem at $t = 0.6$ with 50×50 initial Cartesian grid: final mesh (left); density–distance profile with adaptive method (right).

CRediT authorship contribution statement

Chunyuan Xu: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Conceptualization. **Zhijun Shen:** Writing – review & editing. **Qinghong Zeng:** Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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